Propositional Logic, Knowledge representation, Expert systems.

Book: Discrete Math, Kenneth Rosen, Sections 1.1,1.2,1.3, 1.6, 1.7, 1.8
Russell and Norvig, Sections 7.4 , 7.5

Chris Ding

(For Dr. Ding's class lecture only, not to distribute to others)

- Philosophical Logic
  - 500 BC to 19th Century
- Symbolic Logic
  - Mid to late 19th Century
- Mathematical Logic
  - Late 19th to mid 20th Century
- Logic in Computer Science

- 500 B.C – 19th Century
- Logic dealt with arguments in the natural language used by humans.
- Example
  - All men are mortal.
  - Socrates is a man
  - Therefore, Socrates is mortal.

- **Natural languages are very ambiguous.**
  - Eric does not believe that Mary can pass any test.
    - ...does not believe that she can pass *some* test, or
    - ...does not believe that she can pass *all* tests
  - I only borrowed your car.
    - And not ‘borrowed and used’, or
    - And not ‘car and coat’
  - Tom hates Jim and he likes Mary.
    - Tom likes Mary, or
    - Jim likes Mary
      -
- It leads to many paradoxes.
  - “This sentence is a lie.” (The Liar’s Paradox)

- **Needs a vigours and easy to manipulate logic system**
  - **Propositional logic (Gorge Boole, 1847)**

Logic has profound impact on computer-science

- Some examples:
  - Propositional logic – the foundation of computers and circuitry
  - Databases – query languages
  - Programming languages (e.g. prolog)
  - Design Validation and verification
  - AI (e.g. inference systems)
  - Many other areas

Proposition

A proposition is a declarative sentence that is either true or false.

Which ones of the following sentences are propositions?

- Ottawa is the capital of Canada.
- Buenos Aires is the capital of Brazil.
- $2 + 2 = 4$
- $2 + 2 = 5$
- if it rains, we don't need to bring an umbrella.
- $x + 2 = 4$
- $x + y = z$
- When does the bus come?
- Do the right thing.

Propositional Logic

- Propositional Logic – the area of logic that deals with propositions
- Propositional Variables – variables that represent propositions: *p*, *q*, *r*, *s*
  - E.g. Proposition *p* – “Today is Friday.”
- Truth values – T, F

Propositional Logic: Semantics

- Truth tables define the semantics (=meaning) of the operators
- Convention: 0 = false, 1 = true

p	q	p & q	p or q	p -> q
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	1

Conjuction (logical *and*)

A = “Today is Friday.”

B = “It is raining today.”

conjunction $p \wedge q$ = “Today is Friday and it is raining today.”

Truth Table for Conjunction

A	B	$A \wedge B$
0	0	0
0	1	0
1	0	0
1	1	1

Disjunction (logical *or*)

- Disjunction of A and B is A ∨ B
  - A: Today is Friday.
  - B: It is snowing.
  - A ∨ B: Today is Friday or it is snowing.

- This statement is true if any of the following hold:
  - Today is Friday
  - It is snowing
  - Both
- Otherwise it is false

Truth Table for Disjunction

A	*B*	*A ∨ B*
0	0	0
0	1	1
1	0	1
1	1	1

Negation

Let *p* be a proposition.
The negation of *p*, denoted by ¬*p*, is the statement
"It is not the case that *p*."

- Examples
  - Proposition p = "Today is Friday"

The negation ¬*p* = "It is not the case that *today is Friday*."
In simple English, "Today is not Friday." or "It is not Friday today."

  - Proposition p = "At least 10 inches of rain fell today in Miami"

The negation ¬*p* = "It is not the case that *at least 10 inches of rain fell today in Miami*."
In simple English, "Less than 10 inches of rain fell today in Miami"

Conjuction

Let *p* and *q* be propositions. The *conjunction* of *p* and *q*, $p \wedge q$ = the proposition "*p* and *q*".
The conjunction $p \wedge q$ is true when both *p* and *q* are true. It is false otherwise.

Example

p = "Today is Friday." *q* = "It is raining today."

The conjunction $p \wedge q$ = "Today is Friday and it is raining today."

Implication

A	B	$A \rightarrow B$
0	0	1
0	1	1
1	0	0
1	1	1

- A: A programming homework is due.
- B: It is Tuesday.
- A → B:
  - If a programming homework is due, then it must be Tuesday.
- Is this the same?
  - If it is Tuesday, then a programming homework is due.

The most striking property here is
when A is invalid , A → B is always valid.
This is not easily understood or explainable. But it always works out without any difficulty.

Implication $p \rightarrow q$ has interpreations:

- If p, then q
- p implies q
- if p, q
- p is sufficient for q
- q if p
- q whenever p
- q is necessary for p

Implication $p \rightarrow q$ has interpreations:

"p only if q" says that p cannot be true when q is not true.

"p only if q" expresses the same thing as "if p, then q,"

When p is false, q may be either true or false, because the statement says nothing about the truth value of q.

Be careful not to use "q only if p" to express $p \rightarrow q$

"q unless $\neg p$" means that if $\neg p$ is false, then q must be true.

Converse, Inverse, Contrapositive

Converse of $p \to q$ is $q \to p$

Contrapositive of $p \to q$ is $\neg q \to \neg p$

Inverse of $p \to q$ is $\neg p \to \neg q$

$\neg q \to \neg p$ is equivalent to $p \to q$
contrapositive

$q \to p$ is equivalent to $\neg p \to \neg q$
converse inverse

contrapositive is not equivalent to converse or inverse

Bi-directional implication (Equivalence)

A	B	$A \leftrightarrow B$
0	0	1
0	1	0
1	0	0
1	1	1

- A: You can take the flight.
- B: You have a valid ticket.
- A ↔ B
  - You can take the flight if and only if you have a valid ticket (and vice versa).

Truth table to evaluate compound statement

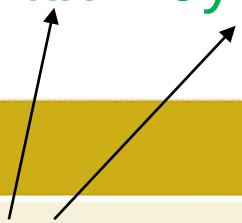
- Truth tables can also be used to determine the truth values of compound statements, such as $(A \vee B) \wedge (\neg A)$ (fill this as an exercise)

A	B	$\neg A$	$A \vee B$	$(A \vee B) \wedge (\neg A)$
0	0	1	0	0
0	1	1	1	1
1	0	0	1	0
1	1	0	1	0

Mathematical expressions have operator precedence

abc+3xy = 4!x + 9y^2

4! = 4*3*2*1



Operators	Precedence
!, +, – (unary operators)	first
*, /, %	second
+, –	third
<, <=, >=, >	fourth
==, !=	fifth
&&	sixth
||	seventh
= (assignment operator)	last

Logical Operator Precedence

abc+3xy = 7! + 9y^2

Precedence of Logical Operators.

Operator		Precedence
$\neg$	(Unary, like x^2， 4！)	1
$\wedge$	(AND multiplication)	2
$\vee$	(OR addition)	3
$\rightarrow$	(like “>“)	4
$\leftrightarrow$	(like “=“)	5

E.g. $\neg p \wedge q = (\neg p) \wedge q$

$p \wedge q \vee r = (p \wedge q) \vee r$

$p \vee q \wedge r = p \vee (q \wedge r)$

Tautology, Contradiction, satisfiable

- A *tautology* is a compound proposition that is always true.
- A *contradiction* is a compound proposition that is always false.
- A *contingency* is neither a tautology nor a contradiction.
- A compound proposition is *satisfiable* if there is at least one assignment of truth values to the variables that makes the statement true.

A	$\neg A$	$A \vee \neg A$	$A \wedge \neg A$
0	1	1	0
1	0	1	0

Result is always true, no matter what A is

Therefore, it is a ***tautology***

Result is always false, no matter what A is

Therefore, it is a ***contradiction***

Logical Equivalence

- Two compound propositions, p and q, are logically equivalent if $p \leftrightarrow q$ is a tautology.
- Notation: $p \equiv q$
- De Morgan's Laws:
- $\neg (p \wedge q) \equiv \neg p \vee \neg q$
- $\neg (p \vee q) \equiv \neg p \wedge \neg q$

proof

p	q	$\neg p$	$\neg q$	$(p \wedge q)$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0

Logical Equivalence

$$p \rightarrow q \equiv \neg p \vee q$$

By definition

proof

p	*q*	¬p	p → q	¬p ∨ q
0	0	1	1	1
0	1	1	1	1
1	0	0	0	0
1	1	0	1	1

Logical Equivalence: Distribution Law

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

proof

p	q	r	$q \wedge r$	$p \vee q$	$p \vee r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	1	0	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

Constructing New Logical Equivalences

- Example: Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

  Solution:

$$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \quad \textit{by example on slide 21}$$
$$\equiv \neg(\neg p) \wedge \neg q \quad \textit{by the second De Morgan law}$$
$$\equiv p \wedge \neg q \quad \textit{by the double negation law}$$

- *Example: Show that* $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to T.

$$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q) \quad \textit{by example on slide 21}$$
$$\equiv (\neg p \vee \neg q) \vee (p \vee q) \quad \textit{by the first De Morgan law}$$
$$\equiv (\neg p \vee p) \vee (\neg q \vee q) \quad \textit{by the associative and communicative law for disjunction}$$
$$\equiv T \vee T$$

Logical Equivalence

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$
$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$
$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$
$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$
$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$
$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$
$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$
$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$
$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$
$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$
$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$
$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

a*(b+c) = a*b + a*c distribution of standard math

Bit Operations

A computer bit has two possible values: 0 (false) and 1 (true). A variable is called a Boolean variable is its value is either true or false.

Bit operations correspond to the logical connectives:

$\vee$ *OR*

$\wedge$ *AND*

$\oplus$ *XOR*

Information can be represented by bit strings, which are sequences of zeros and ones, and manipulated by operations on the bit strings.

Proof and Induction

Proof methods

Proof methods for P → Q

Exhaustive proof: demonstrate for all possible cases

Direct proof: Assume P, prove Q

Proof by contradiction: Assume $P \wedge \neg Q$; show it is false

Proof by contraposition: assume $\neg Q$, prove ¬P

Direct Proof example

Constructing New Logical Equivalences

- Example: Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

  Solution:

$$\begin{aligned} \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \textit{by example on slide 21} \\ &\equiv \neg(\neg p) \wedge \neg q && \textit{by the second De Morgan law} \\ &\equiv p \wedge \neg q && \textit{by the double negation law} \end{aligned}$$

- *Example: Show that* $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to T.

$$\begin{aligned} (p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \textit{by example on slide 21} \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \textit{by the first De Morgan law} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) && \textit{by the associative and} \\ & && \textit{communicative law for disjunction} \\ &\equiv T \vee T \end{aligned}$$

Most widely used inference rules

- Inference rules:

$$\frac{\text{Antecedents}}{\text{Consequent}} \quad \text{(rule-name)}$$

- Examples:

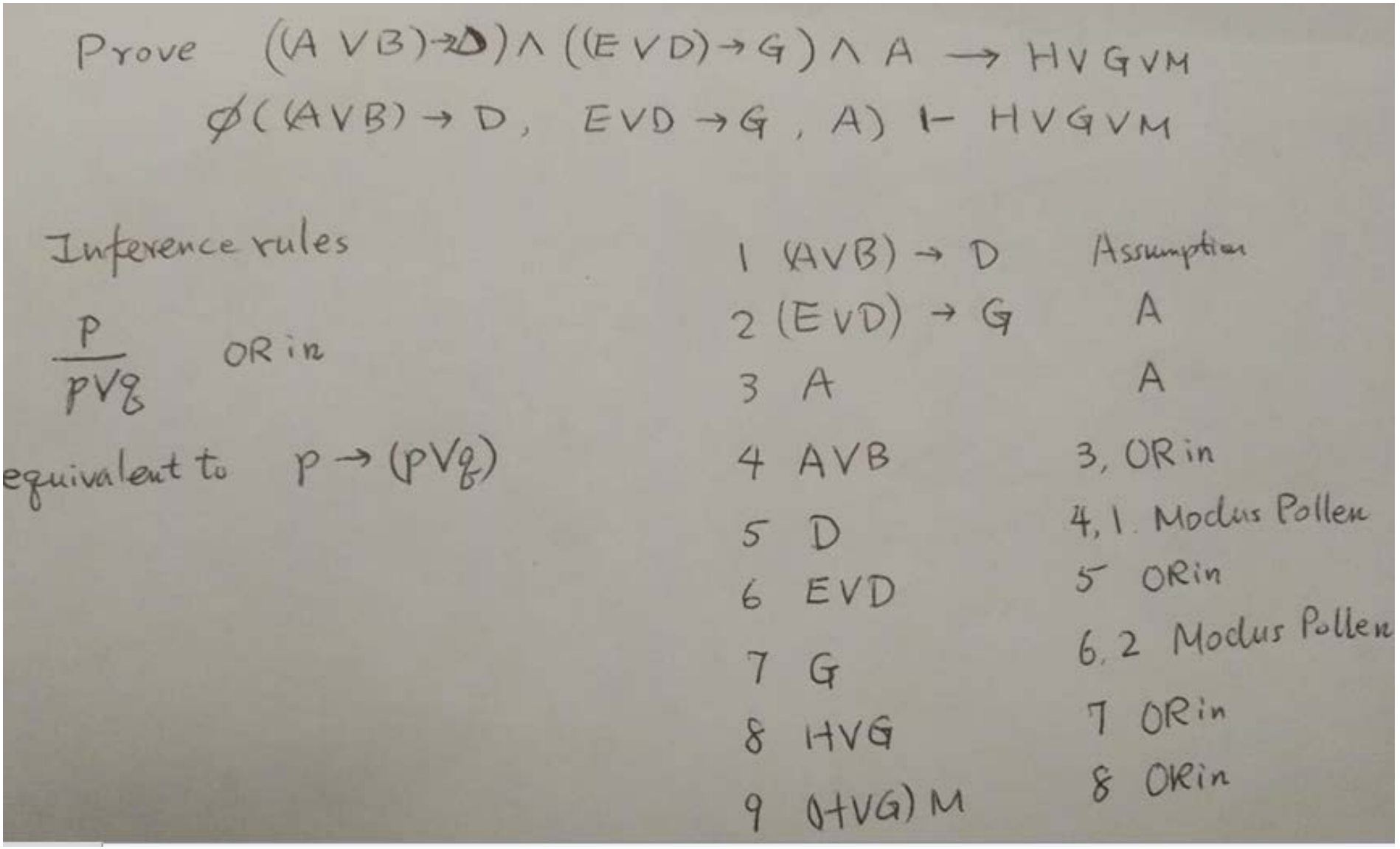
$$\frac{A \rightarrow B \quad B \rightarrow C}{A \rightarrow C} \quad \text{(transitive)}$$

$$\frac{A \rightarrow B \quad A}{B} \quad \text{(Modus Pollen)}$$

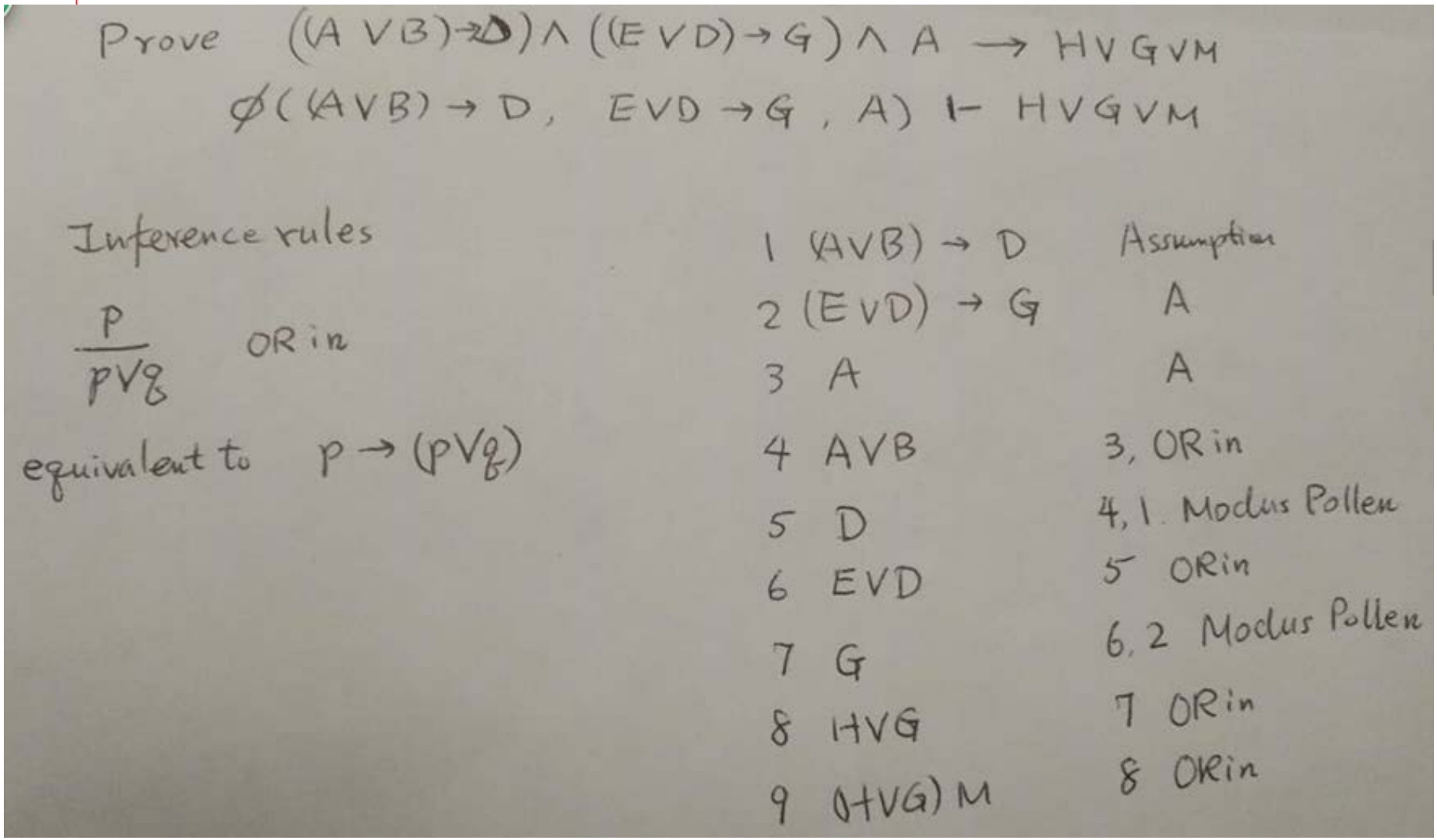
$$\frac{A \rightarrow B \quad \sim B}{\sim A} \quad \text{(Modus Tollens)}$$

$$\emptyset(A \rightarrow B, B \rightarrow C) \qquad A \rightarrow B$$

Truth Tree Proof Example



Truth Tree Proof Example



Prove

$$\phi(S \to ((H \to M) \to R),\ (M \to (H \to M)) \to ((H \to M) \to S),\ M,\ M \to (H \to M)) \to R$$

Proof

1.	$S \to ((H \to M) \to R)$	
2.	$(M \to (H \to M)) \to ((H \to M) \to S)$	
3.	M	
4.	$M \to (H \to M)$	
5.	$(H \to M) \to S$	2, 4
6.	$H \to M$	3, 4
7.	S	5, 6
8.	$(H \to M) \to R$	7, 1
9	R	6, 8

Propositional Logic Resolution

The resolution inference

In the form of clauses (CNF)

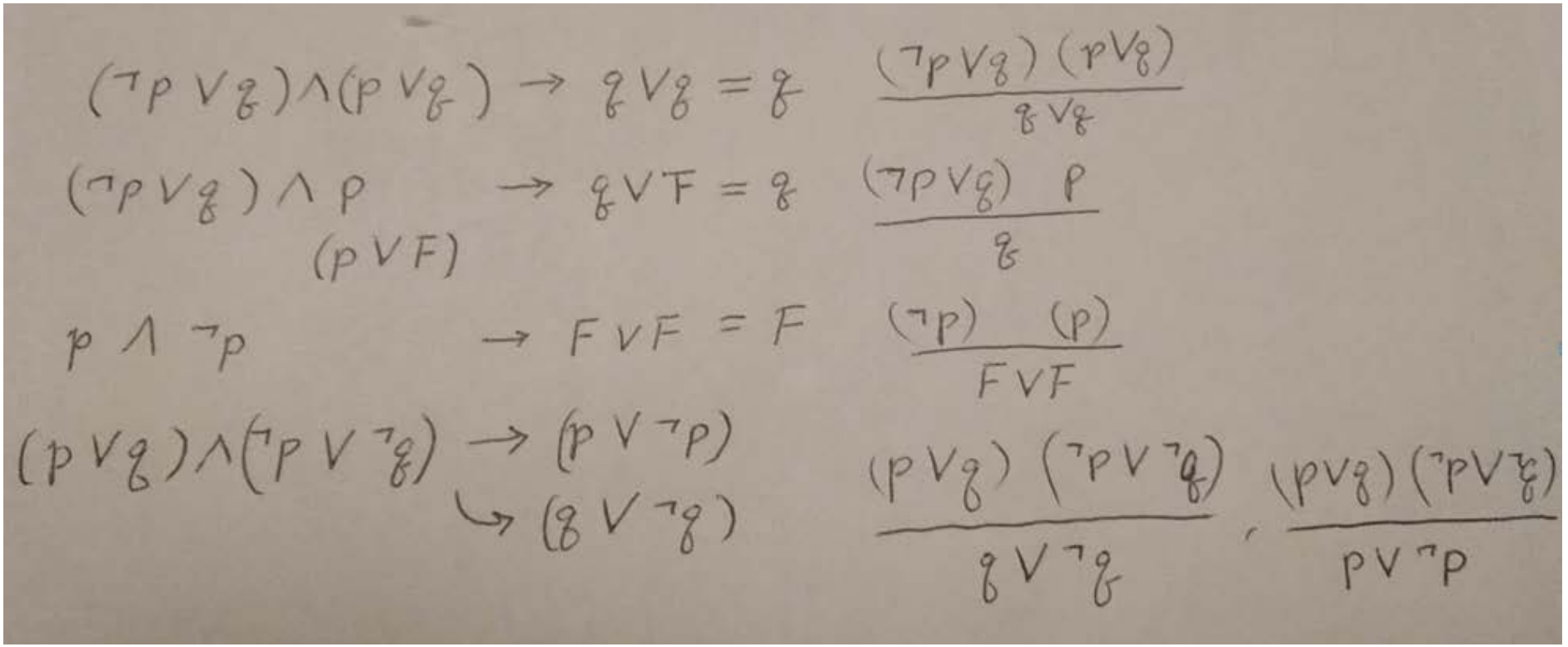
$$\frac{(l \vee l_1 \vee .. \vee l_n) \qquad (\neg l \vee l'_1 \vee .. \vee l'_n)}{(l_1 \vee .. \vee l_n \vee l'_1 \vee .. \vee l'_n)} \text{ (Resolution)}$$

Example:

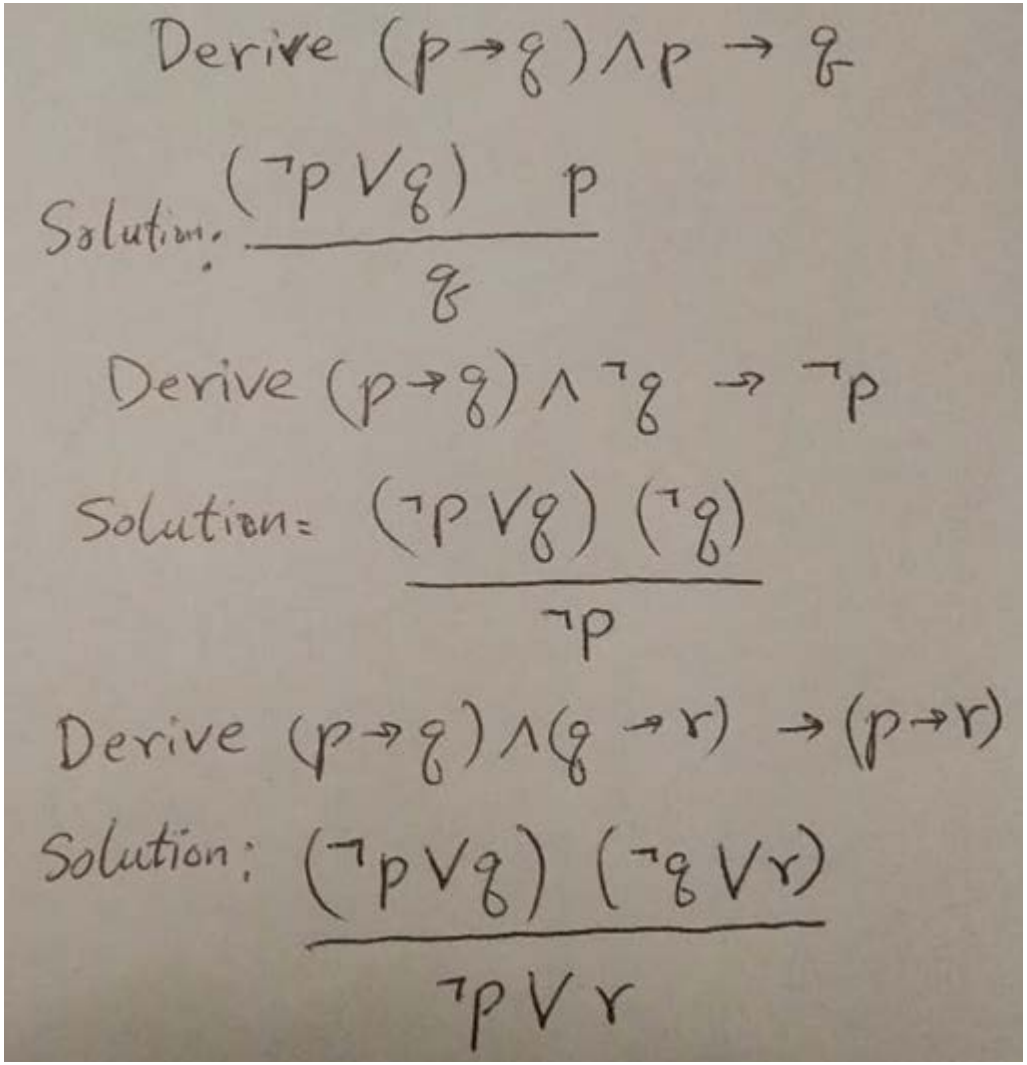
$$\frac{(a \vee b) \qquad (\neg a \vee c)}{(b \vee c)}$$

Homework: Prove: $(a \vee b) \wedge (\neg a \vee c) \rightarrow (b \vee c)$

Propositional Logic Resolution



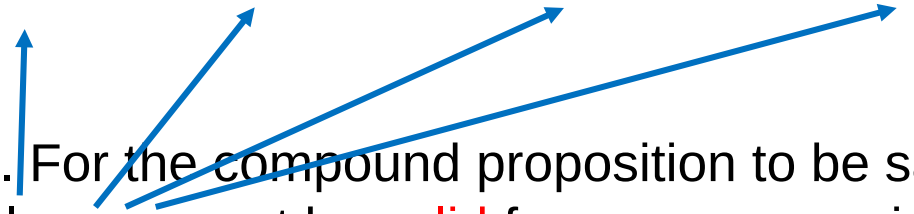
Propositional Logic Resolution



Use truth table to prove it

Use resolution to prove that

(p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.



Proof. For the compound proposition to be satisfiable, each of the four clauses must be valid for a proper assignment of p,q,r. Thus by resolution, we have

By distribution, A&B&C&D = D&A&B&C
Thus the above compound proposition becomes

（p V ~q) & (p V q) & (~p V r) & (~p V ~r)
= [(p V ~q) & (p V q)] & [(~p V r) & (~p V ~r)]
= [p V p] & [~p V ~p] do resolution on both
= [p] & [~p] why?
= false why?

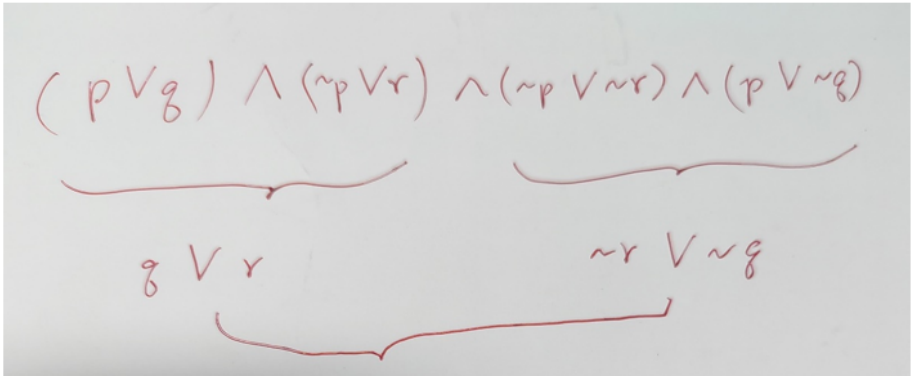
Thus, there is no assignment that makes the proposition true. It is unsatisfiable.
(We start with the assumption that an assignment makes the compound proposition true. We use resolution inference to show the proposition is false.)

This is another attempt to prove. It did not provide useful things.

Use resolution to prove that

(p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.

Proof. For the compound proposition to be satisfiable, each of the four clauses must be valid for a proper assignment of p,q,r. Thus by resolution, we have



r V ~r = true

We start with the whole compound resolution is true. We end up with it is true. We did not prove/get anything new.

Propositional Logic Proof Technique:

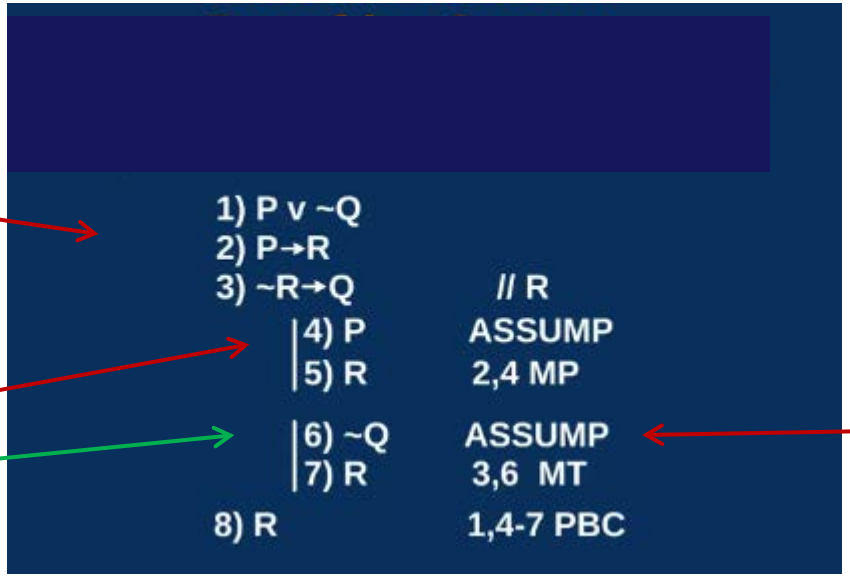
Assume of the variable is true,
prove all choices are consistently true

Divide the assumption into different cases, prove a compound proposition

$\varnothing$ (P v ˜Q, P→ R, ˜R → Q) → R (Proposition A)

The three conditions

Because P V ˜Q is true.
Thus (i) either P is true,
or (ii) ~Q is true.



Contrapositive of (3) is ~Q → R

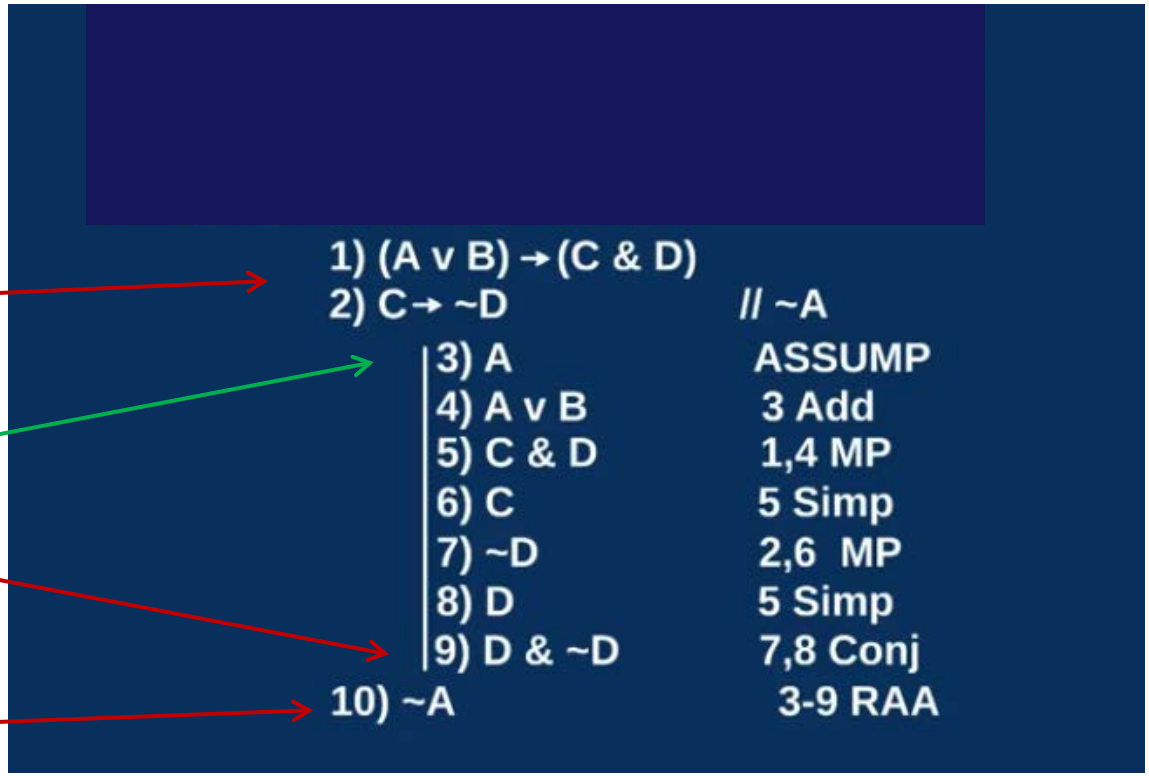
In both cases, R is true. Thus, the proposition is true

Homework:

Prove the same proposition using (i) truth tree and (ii) resolution methods.

Prove by contradiction

$$\varnothing\ (\ (A \vee B) \rightarrow (C \ \&\ D),\ C \rightarrow \sim D\ ) \rightarrow \sim A$$



Two conditions

Assume A is true.

This is a contradiction.
Thus the assumption
(A is true) can not hold.
Thus ~A holds.

Homework: use truth tree to prove the same proposition

$$Ø(H \rightarrow D,\ U \rightarrow S) \rightarrow ((H \,\&\, U) \rightarrow (S \,\&\, D))$$

We assume H & U is true.

Using two preconditions, we obtain that S & D is true.

Thus (H & U) → (S & D) is true.

So, we proved that in A → B, B is true. This is sufficient for A → B is true (why?).

1) H→D	
2) U→S	// (H & U)→(S & D)
3) H & U	ASSUMP
4) H	3 Simp
5) D	1,4 MP
6) U	3, Simp
7) S	2,6 MP
8) S & D	7,5 Conj
9) (H&U)→(S&D)	3-8 CP

Homework:

(a) Prove: $(a \vee b) \wedge (\neg a \vee c) \rightarrow (b \vee c)$

(b)



2. *Propositional Resolution.* Use propositional resolution to show that the following sets of clauses are unsatisfiable.
   (a) $\{p,q\}, \{\neg p,r\}, \{\neg p,\neg r\}, \{p, \neg q\}$
   (b) $\{p,q, \neg r,s\}, \{\neg p,r,s\}, \{\neg q, \neg r\}, \{p, \neg s\}, \{\neg p, \neg r\}, \{r\}$

Note {p,q} = p V q is a clause. We use "~" for negation operator, "&" for AND operator. Thus (a) means show (p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.

(c)

7.17 A propositional *2-CNF* expression is a conjunction of clauses, each containing *exactly 2* literals, e.g.,

$$(A \vee B) \wedge (\neg A \vee C) \wedge (\neg B \vee D) \wedge (\neg C \vee G) \wedge (\neg D \vee G) .$$

a. Prove using resolution that the above sentence entails G.

"entail" = "imply"

Homework:

(d)

7.10 Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables or the equivalence rules

a. $Smoke \Rightarrow Smoke$

b. $Smoke \Rightarrow Fire$

c. $(Smoke \Rightarrow Fire) \Rightarrow (\neg Smoke \Rightarrow \neg Fire)$

d. $Smoke \vee Fire \vee \neg Fire$

e. $((Smoke \wedge Heat) \Rightarrow Fire) \Leftrightarrow ((Smoke \Rightarrow Fire) \vee (Heat \Rightarrow Fire))$

f. $(Smoke \Rightarrow Fire) \Rightarrow ((Smoke \wedge Heat) \Rightarrow Fire)$

g. $Big \vee Dumb \vee (Big \Rightarrow Dumb)$

(c)

A large number of real world applications deal with satisfiability problem

Propositional Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true. When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**.

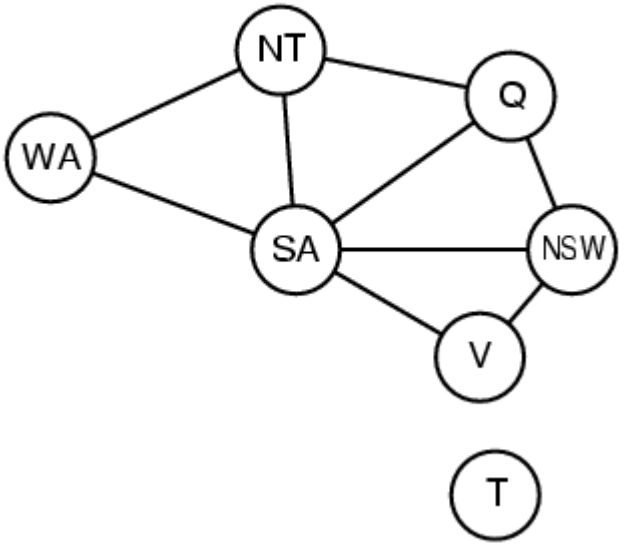
Is compound proposition $x \vee y \rightarrow \neg x \wedge z$ satisfiable ?

	x	y	z	$x \vee y \rightarrow \neg x \wedge z$
1	F	F	F	T
2	F	F	T	T
3	F	T	F	F
4	F	T	T	T
5	T	F	F	F
6	T	F	T	F
7	T	T	F	F
8	T	T	T	F

SAT Problem

- Find a coloring of Australia map using 2 colors (T, F).





$(WA \neq NT) \ \& \ (WA \neq SA) \ \& \ (NT \neq SA) \ \& \ (NT \neq Q) \cdots\cdots$

Truth Tree

1. Proof technique
2. Check satisfiability

Using counter example/
Check satisfiability

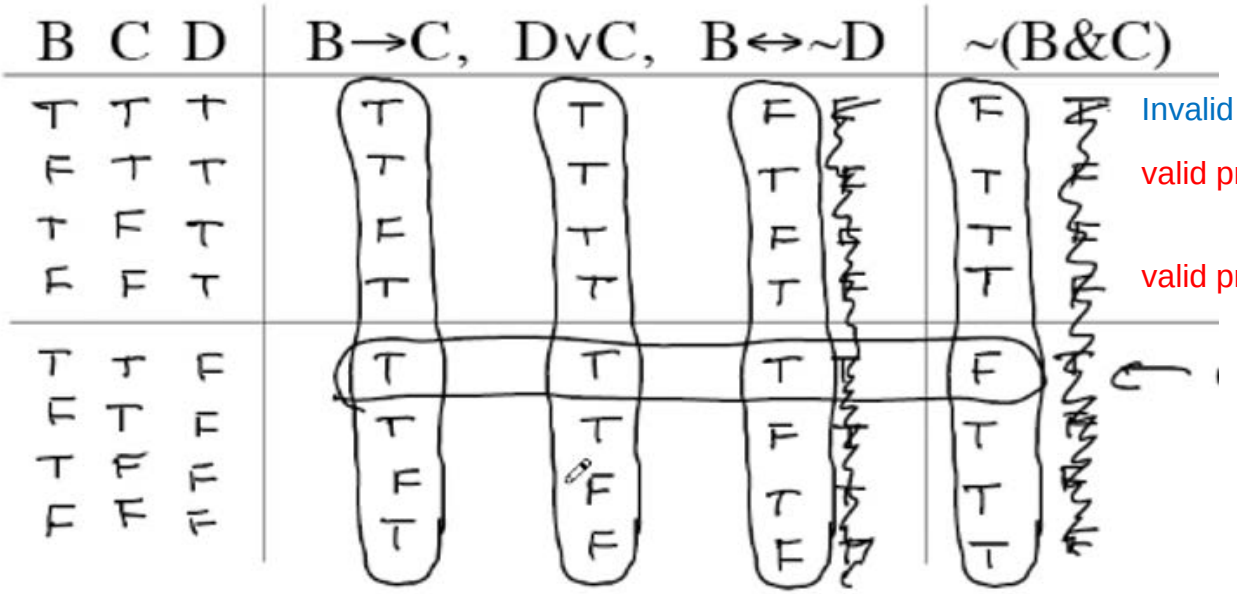
We wish to prove: Collection of premises/assumptions/conditions → consequence, conclusion

$$\emptyset(B\rightarrow C,\ D\vee C,\ B\leftrightarrow\neg D) \rightarrow \neg B\&C$$

Check the validity
(Is this always true ?)

$$(B\rightarrow C)\wedge(D\vee C)\wedge(B\leftrightarrow\neg D) \rightarrow \neg(B\&C)$$

Truth table:



Invalid premisis, statement is always true

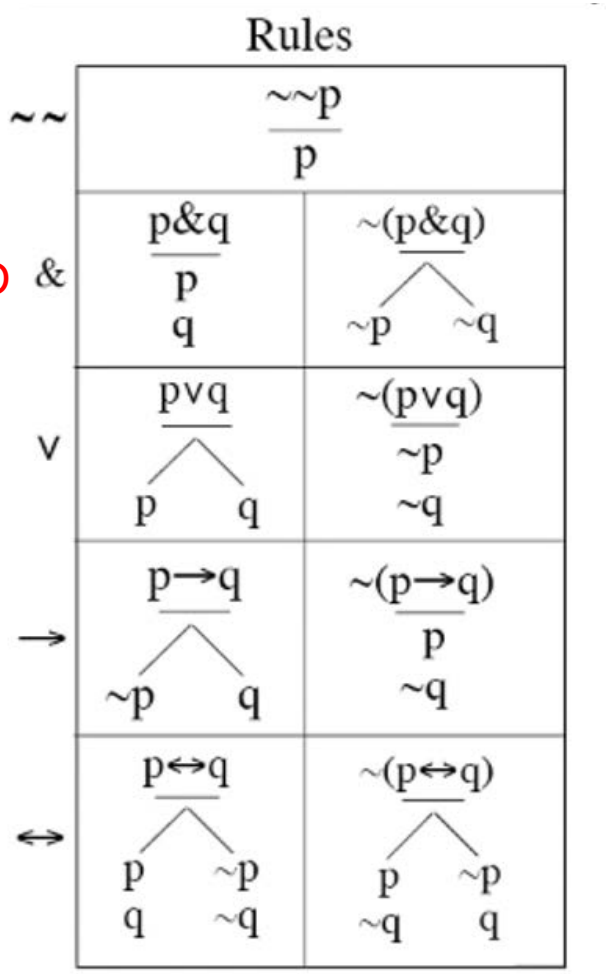
valid premisis, valid consequence

valid premisis, valid consequence

valid premisis, consequence false

There are 3 cases with valid premisis.
The consequence is true in 2 cases, failed in 1 case.
THUS, the statement is false, invalid.
Truth table computation is slow. Can we do faster?

Given something is true, we can deduct consequences:
This is truth tree induction



~p V q

~p V ~q

~p AND ~q

p AND ~q

We check the validity (satisfiability) of

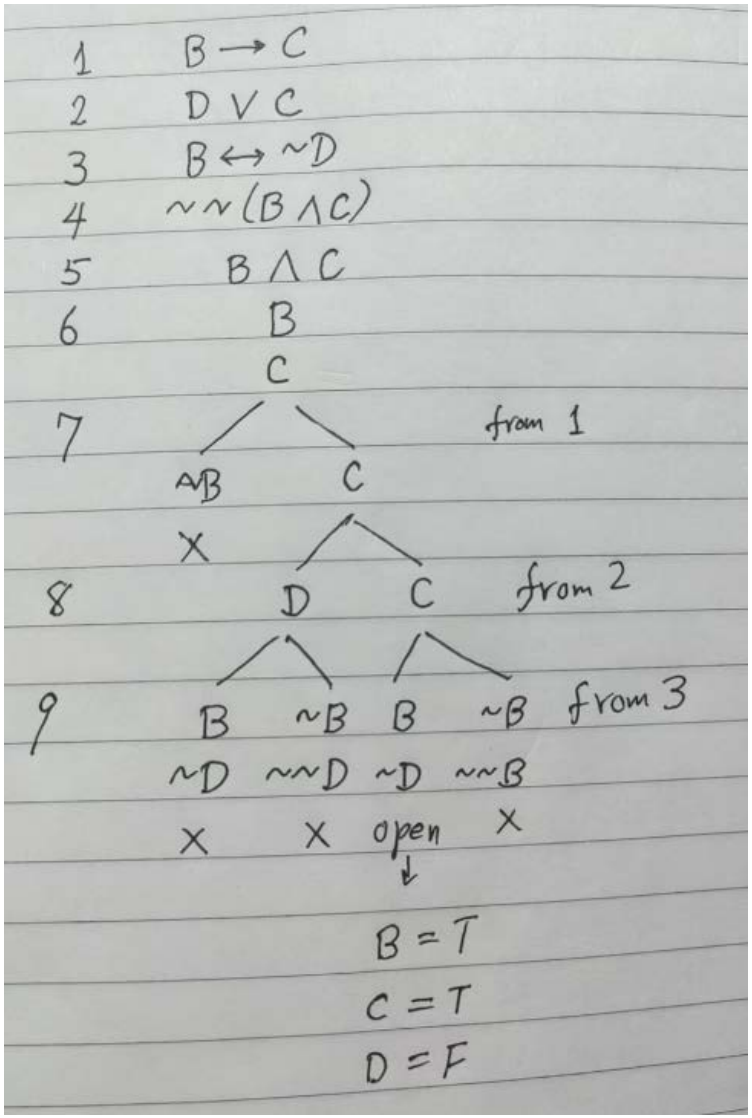
add negation

$$(B \rightarrow C) \wedge (D \vee C) \wedge (B \leftrightarrow \sim D) \wedge (\sim \sim (B \& C))$$

Truth Trees for Validity

Rules

~~	~~p p	
&	p&q p q	~(p&q) ~p ~q
v	pvq p q	~(pvq) ~p ~q
→	p→q ~p q	~(p→q) p ~q
↔	p↔q p ~p q ~q	~(p↔q) p ~p ~q q



Outine of advanced logical forms and NP-hardness

Satisfiability

Negation normal form

Conjunctive normal form (CNF) = product of sums

Disjunctive normal form (DNF) = sum of products

Propositional Satisfiability can be NP-hard

Horn Clause --- decision rules

Expert Systems

Propositional Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true. When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**.

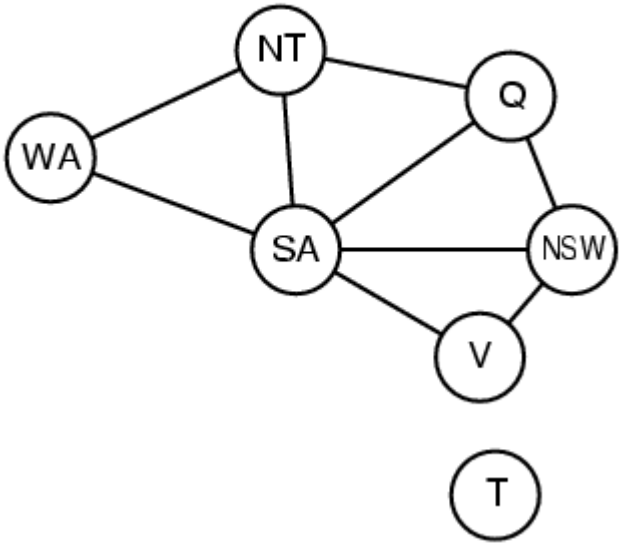
Is compound proposition $x \vee y \rightarrow \neg x \wedge z$ satisfiable ?

	x	y	z	$x \vee y \rightarrow \neg x \wedge z$
1	F	F	F	T
2	F	F	T	T
3	F	T	F	F
4	F	T	T	T
5	T	F	F	F
6	T	F	T	F
7	T	T	F	F
8	T	T	T	F

SAT Problem

- Find a coloring of Australia map using 2 colors (T, F).





$$(WA \neq NT) \,\&\, (WA \neq SA) \,\&\, (NT \neq SA) \,\&\, (NT \neq Q) \cdots\cdots$$

Can we find ONE assignments of WA, NT, …W,
so that the above propersition is true?

Satisfiability is a key technology in AI and computer science

Homework: use truth table to show the three compound statements here are satisfiable.

EXAMPLE 9 (from Rosen Book, page 31)

Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

Solution: Instead of using truth table to solve this problem, we will reason about truth values. Note that $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when the three variable p, q, and r have the same truth value (see Exercise 40 of Section 1.1). Hence, it is satisfiable as there is at least one assignment of truth values for p, q, and r that makes it true. Similarly, note that $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p, q, and r is true and at least one is false (see Exercise 41 of Section 1.1). Hence, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable, as there is at least one assignment of truth values for p, q, and r that makes it true.

Finally, note that for $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ to be true, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ and $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ must both be true. For the first to be true, the three variables must have the same truth values, and for the second to be true, at least one of three variables must be true and at least one must be false. However, these conditions are contradictory. From these observations we conclude that no assignment of truth values to p, q, and r makes $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ true. Hence, it is unsatisfiable. ◀

Satisbility problems (Rosen, p.36)

58. How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to p, q, and r?

59. How many of the disjunctions $p \vee \neg q \vee s$, $\neg p \vee \neg r \vee s$, $\neg p \vee \neg r \vee \neg s$, $\neg p \vee q \vee \neg s$, $q \vee r \vee \neg s$, $q \vee \neg r \vee \neg s$, $\neg p \vee \neg q \vee \neg s$, $p \vee r \vee s$, and $p \vee r \vee \neg s$ can be made simultaneously true by an assignment of truth values to p, q, r, and s?

60. Show that the negation of an unsatisfiable compound proposition is a tautology and the negation of a compound proposition that is a tautology is unsatisfiable.

Satisbility problems (Rosen, p.36)

61. Determine whether each of these compound propositions is satisfiable.

a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$

b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$

c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

62. Determine whether each of these compound propositions is satisfiable.

a) $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s) \wedge (p \vee q \vee \neg s)$

b) $(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg s) \wedge (\neg p \vee \neg r \vee \neg s) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg r \vee \neg s)$

c) $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg s) \wedge (q \vee \neg r \vee s) \wedge (\neg p \vee r \vee s) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee s) \wedge (\neg p \vee \neg r \vee \neg s)$

A collection of logical operators is called **functionally complete** if every compound proposition is logically equivalent to a compound proposition involving only these logical operators.

43. Show that $\neg$, $\wedge$, and $\vee$ form a functionally complete collection of logical operators. [*Hint:* Use the fact that every compound proposition is logically equivalent to one in disjunctive normal form, as shown in Exercise 42.]

Negation Normal Form (NNF)

- Definition: A formula is said to be in Negation Normal Form (NNF) if it only contains $\neg$, $\wedge$ and $\vee$ connectives and only atoms can be negated.

- Examples:
  - $\phi_1 = \neg(A \vee \neg B)$ is not in NNF
  - $\phi_2 = \neg A \wedge B$ is in NNF

$\neg$, $\wedge$, and $\vee$ form a functionally complete collection of logical operators.

Converting to NNF

- Every formula can be converted to NNF in linear time:
  - Eliminate all connectives other than $\wedge$, $\vee$, $\neg$
  - Use De Morgan and double-negation rules to push negations to the right
- Example: $\phi = \neg(A \rightarrow \neg B)$
  - Eliminate '$\rightarrow$': $\phi = \neg(\neg A \vee \neg B)$
  - Push negation using De Morgan: $\phi = (\neg\neg A \wedge \neg\neg B)$
  - Use Double negation rule: $\phi = (A \wedge B)$

Normal forms for compound propositions

- A literal is a propositional variable or the negation of a propositional variable.
- A term is a literal or the conjunction (and) of two or more literals.
- A clause is a literal or the disjunction (or) of two or more literals.

Definition

A compound proposition is in **disjunctive normal form** (DNF) if it is a term or a disjunction of two or more terms. (sum of products)
A compound proposition is in **conjunctive normal form** (CNF) if it is a clause or a conjunction of two or more clauses. (product of sums)

CNF: $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s)$ (1+x)* (xy+z)*(3x+w)

DNF: $(\neg p \wedge \neg q) \vee (q \wedge p)$ 3 + xy + xz^2 + 5xw

Conversion between DNF and CNF $(\neg p \wedge \neg q) \vee (q \wedge p) = (\neg p \vee q) \wedge (\neg q \vee p)$

Literals, terms, clauses

- Definition: A **literal** is either a proposition or the negation of a proposition

  Example: A, ¬**B**

- Definition: a term is a conjunction of literals
  - Example: $(A \wedge \neg B \wedge C)$

- Definition: a clause is a disjunction of literals
  - Example: $(A \vee \neg B \vee C)$

Disjunctive Normal Form (DNF)

- Definition: A formula is said to be in Disjunctive Normal Form (DNF) if it is a disjunction of terms.
  - In other words, it is a formula of the form

$$\bigvee_i \left(\bigwedge_j l_{i,j} \right)$$

where $l_{i,j}$ is the j-th literal in the i-th term.

In standard math: sums of terms $3 + xy + xz^2 + 5xw$

- Examples
  - $\phi = (A \wedge \neg B \wedge C) \vee (\neg A \wedge D) \vee (B)$ is in DNF

Conjunctive Normal Form (CNF)

- Definition: A formula is said to be in Conjunctive Normal Form (CNF) if it is a conjunction of clauses.
  - In other words, it is a formula of the form

$$\bigwedge_i \left(\bigvee_j l_{i,j} \right)$$

where $l_{i,j}$ is the j-th literal in the i-th term.

In standard math: product of sums (1+x)* (xy+z)*(3x+w)

- Examples
  - $\phi = (A \vee \neg B \vee C) \wedge (\neg A \vee D) \wedge (B)$ is in CNF

De Morgan's negation on CNF (not on literals)

(A & ~B & C) v (~A & D) v (B) in DNF

$$\phi = A$$

$$\phi = C \wedge \sim B$$

$$\phi = A \vee \sim B \vee C$$

$$\phi = (A \wedge \sim B \wedge C) \vee (\sim A \wedge D) \vee B$$

- Every formula can be converted to DNF in exponential time and space:
  - Convert to NNF
  - Distribute disjunctions following the rule:
    $\models A \wedge (B \vee C) \leftrightarrow ((A \wedge B) \vee (A \wedge C))$
- Example:
  - $\phi = (A \vee B) \wedge (\neg C \vee D) =$
    $((A \vee B) \wedge (\neg C)) \vee ((A \vee B) \wedge D) =$
    $(A \wedge \neg C) \vee (B \wedge \neg C) \vee (A \wedge D) \vee (B \wedge D)$
  - Q: how many clauses would the DNF have had we started from a conjunction of **n** clauses ?

Direct Proof example

CNF to DNF

$(p \rightarrow q) \wedge (q \rightarrow p)$

$= (\neg p \vee q) \wedge (\neg q \vee p)$

$= (\neg p \vee q) \wedge (\neg q \vee p)$ (use distribution to obtain)

$= ((\neg p \vee q) \wedge \neg q) \vee ((\neg p \vee q) \wedge p)$ (use distribution to obtain)

$= (\neg p \wedge \neg q) \vee (q \wedge \neg q) \vee (\neg p \wedge p) \vee (q \wedge p)$

$= (\neg p \wedge \neg q) \vee \bot \vee \bot \vee (q \wedge p)$

$= (\neg p \wedge \neg q) \vee (q \wedge p)$

Disjunctive normal form (DNF)

	x	y	z	$x \vee y \rightarrow \neg x \wedge z$
1	F	F	F	T
2	F	F	T	T
3	F	T	F	F
4	F	T	T	T
5	T	F	F	F
6	T	F	T	F
7	T	T	F	F
8	T	T	T	F

The formula is satisfied by the truth assignment in row 1 **or**
by the truth assignment in row 2 **or** by the truth assignment in row 4.
So, its DNF is : $(\neg x \wedge \neg y \wedge \neg z) \vee (\neg x \wedge \neg y \wedge z) \vee (\neg x \wedge y \wedge z)$

Conjunctive normal form (CNF)

	x	y	z	$x \vee y \rightarrow \neg x \wedge z$
1	F	F	F	T
2	F	F	T	T
3	F	T	F	F
4	F	T	T	T
5	T	F	F	F
6	T	F	T	F
7	T	T	F	F
8	T	T	T	F

Cannot be this

The formula is **not** satisfied by the truth assignment in row 3 **and** in row 5 **and** in row 6 **and** in row 7 **and** in row 8. So:, it is log. equiv. to:

$$\neg(\neg x \wedge y \wedge \neg z) \wedge \neg(x \wedge \neg y \wedge \neg z) \wedge \neg(x \wedge \neg y \wedge z) \wedge \neg(x \wedge y \wedge \neg z) \wedge \neg(x \wedge y \wedge z)$$

apply DeMorgan's law to obtain its CNF:

$$(x \vee \neg y \vee z) \wedge (\neg x \vee y \vee z) \wedge (\neg x \vee y \vee \neg z) \wedge (\neg x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee \neg z)$$

Propositional satisfiabilty is NP-hard

Is your formula in CNF or DNF? - In satisfiability, the constraints are specified as a logical formula. There are two primary ways of expressing logical formulas, conjunctive normal form (CNF) and disjunctive normal form (DNF). In CNF formulas, we must satisfy all clauses, where each clause is constructed by and-ing or's of literals together, such as

$$(v_1 \text{ or } \bar{v}_2) \text{ and } (v_2 \text{ or } v_3)$$

All conditions must be true

In DNF formulas, we must satisfy any one clause, where each clause is constructed by or-ing ands of literals together. The formula above can be written in DNF as

One condition mts be true

$$(\bar{v}_1 \text{ and } \bar{v}_2 \text{ and } v_3) \text{ or } (\bar{v}_1 \text{ and } v_2 \text{ and } \bar{v}_3) \text{ or } (\bar{v}_1 \text{ and } v_2 \text{ and } v_3) \text{ or } (v_1 \text{ and } \bar{v}_2 \text{ and } v_3)$$

Solving DNF-satisfiability is trivial, since any DNF formula can be satisfied unless *every* clause contains both a literal and its complement (negation). However, CNF-satisfiability is NP-complete. This seems paradoxical, since we can use De Morgan's laws to convert CNF-formulae into equivalent DNF-formulae and vice versa. The catch is that an exponential number of terms might be constructed in the course of translation, so that the translation itself might not run in polynomial time.

Propositional satisfiabilty on CNF

Suppose the system has n variables

2SAT problem

$(x_1 \vee x_3) \wedge (x_2 \vee \neg x_8) \wedge (x_5 \wedge \neg x_7) \wedge (x_4 \wedge x_7) \wedge \cdots\cdots$

3SAT problem

$(x_1 \vee x_2 \vee x_5) \wedge (x_2 \vee x_7 \vee x_{11}) \wedge (\neg x_3 \vee x_9 \vee x_{11}) \wedge \cdots\cdots$

2SAT problem can be solved in linear time

3SAT problem has no polynomial time algorithm: NP-hard (Cook, 1970)

Homework:
Exercises 27, 30, 31 on p.15 (see below)

27. State the converse, contrapositive, and inverse of each of these conditional statements.

a) If it snows today, I will ski tomorrow.
b) I come to class whenever there is going to be a quiz.
c) A positive integer is a prime only if it has no divisors other than 1 and itself.

30. How many rows appear in a truth table for each of these compound propositions?

a) $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
b) $(p \vee \neg t) \wedge (p \vee \neg s)$
c) $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
d) $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$

31. Construct a truth table for each of these compound propositions.

a) $p \wedge \neg p$
b) $p \vee \neg p$
c) $(p \vee \neg q) \rightarrow q$
d) $(p \vee q) \rightarrow (p \wedge q)$
e) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
f) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

Homework:

Use truth table to prove the following three compound propositions.

EXAMPLE 9 (from Rosen Book, page 31)

Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

Homework:

In standard mathematics, can the following three equations hold (being true) simultaneously?

$$x = y$$
$$y = z$$
$$z = -x$$

Why?

In propositional logic, consider the following three compound propositions

$$A \Rightarrow B$$
$$B \Rightarrow C$$
$$C \Rightarrow \neg A$$

(a) They can not be true simultaneously in all cases?
(b) They can be true simultaneously in some cases?
(c) They can be true simultaneously in all cases?
Provide your answer. Explain or prove it.
(Hint: use the truth table to guide your thinking.)

Horn clause

Definition: A literal is *positive* if is a proposition name. A literal is *negative* if it is the complement of a proposition name.

Definition: A *Horn clause* is a clause which contains at most one positive literal. The general format of such a clause is thus as follows:

$$\neg A_1 \vee \neg A_2 \vee .. \vee \neg A_n \vee B$$

This may be rewritten as an implication:

$$(A_1 \wedge A_2 \wedge .. \wedge A_n) \rightarrow B \qquad <= \text{ rule}$$

AI is about making decisions, i.e., rules.
Horn clause is the most important clause for AI.

Experts Systems

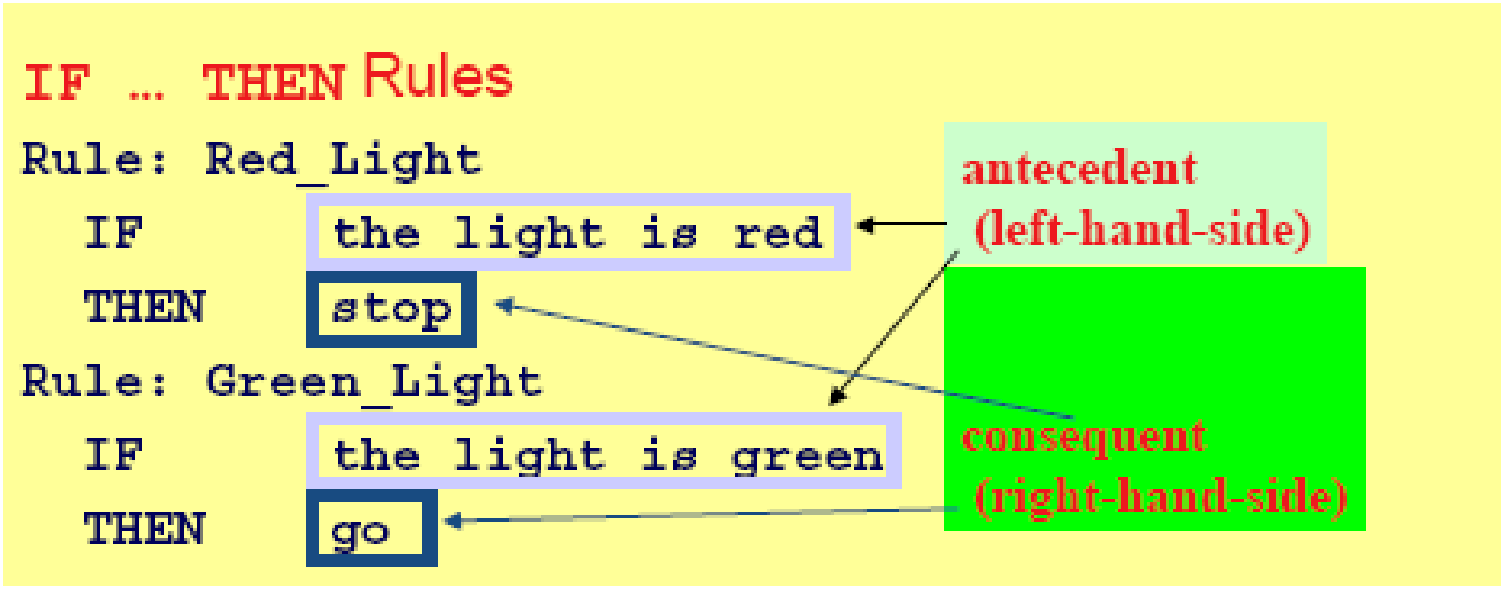
- **A medical diagnose system to check if you have cancer**
- **Anti-lock Braking Systems**
- **Video CAMcorders**
- **Appliances**
  - **Washers**
  - **Toasters**
  - **Stoves**
- **User feedback/Help Desk Software**
- **Subway Control**
- **Data Analysis/Mining Software**

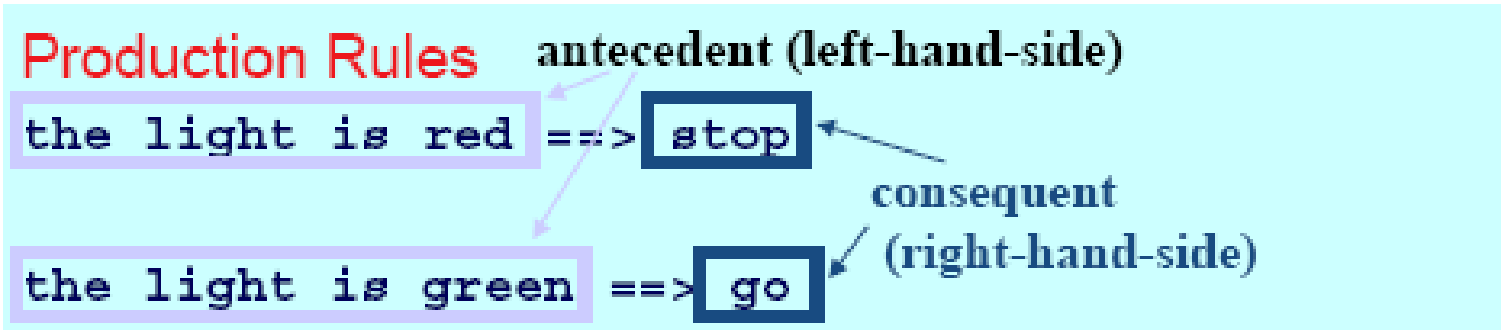
Rule-based expert system (a collection of rules and actions)
Knowledge/facts are represented by rules.

Rule-Based ES

- knowledge is encoded as **`IF ... THEN`** rules
  - these rules can also be written as *production rules*
- the inference engine determines which rule antecedents are satisfied
  - the left-hand side must "match" a fact in the working memory
- satisfied rules are placed on the agenda
- rules on the agenda can be activated ("fired")
  - an activated rule may generate new facts through its right-hand side
  - the activation of one rule may subsequently cause the activation of other rules

Example Rules

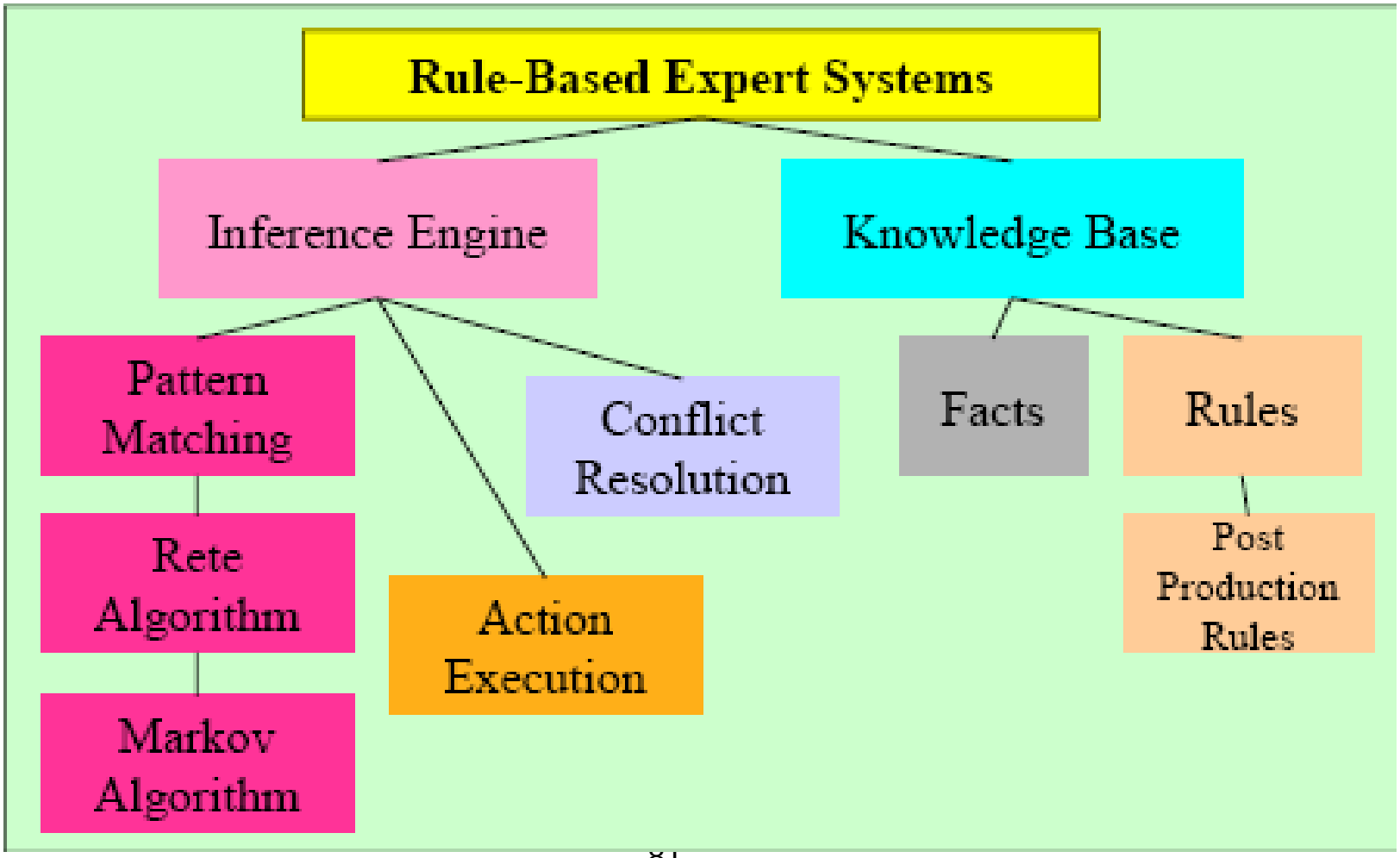




Inference Engine Cycle

- The inference engine determines the execution of the rules by the following cycle:
  - conflict resolution
    - select the rule with the highest priority from the agenda
  - execution (Act)
    - perform the actions on the consequent of the selected rule
    - remove the rule from the agenda
  - match
    - update the agenda
      - add rules whose antecedents are satisfied to the agenda
      - remove rules with non-satisfied agendas
- the cycle ends when no more rules are on the agenda, or when an explicit stop command is encountered

Foundation of Expert Systems



General Methods of Inferencing

- Forward chaining (data-driven)– reasoning from facts to the conclusions resulting from those facts – best for prognosis, monitoring, and control.
  - Examples: CLIPS, OPS5
- Backward chaining (query/Goal driven)– reasoning in reverse from a hypothesis, a potential conclusion to be proved to the facts that support the hypothesis – best for diagnosis problems.
  - Examples: MYCIN

Production Systems

- Rule-based expert systems – most popular type today.
- Knowledge is represented as multiple rules that specify what should/not be concluded from different situations.
- Forward chaining – start w/facts and use rules do draw conclusions/take actions.
- Backward chaining – start w/hypothesis and look for rules that allow hypothesis to be proven true.

Expert System

- Extensive, task-specific knowledge acquired from training, reading and experience
  - Theories about the problem area
  - Hard-and-fast rules and procedures
  - Rules (heuristics)
  - Global strategies
  - Meta-knowledge (knowledge about knowledge)
  - Facts
- Knowledge are represented as rules and procedures

Rule-based Expert system (a collection of rules and actions)

Rule-based Expert system

R1: R1 is a rule-based expert system which was designed by Digital Equipment Corporation (DEC) in the late 1970's for configuration of VAX computer systems.

Examples of a rule from R1:

If:

1. The most current active component is distributing massbus devices.
2. There is a single-port disk drive that has not been assigned to a massbus.
3. There are no unassigned single-port disk drives
4. The number of devices that each massbus should support is known.
5. There is a massbus that has been assigned at least one disk drive and that should support additional drives.
6. The type of cable needed to connect the disk drive to the previous device on the massbus is known.

Then:

- Assign the disk drive to the massbus.

Unlike MYCIN, R1 was a commercial success, partly because the domain was sufficiently restricted.

MYCIN: MYCIN is an expert system which was designed to assist physicians in the diagnosis of and prescription of treatment for infectious blood disease.

Example of a rule from MYCIN:

If:

1. The site of the culture is blood.
2. The stain of the organism is gramneg.
3. The morphology of the organism is rod.
4. The aerobicity of the organism is anerobic.
5. The portal of entry of the organism is GI.

Then:

- There is strongly suggestive evidence (0.9) that the identity of the organism is bacteroides.

MYCIN is the "grandaddy" of all rule-based expert systems, but was never used in a practical clinical setting because of its *brittleness.*

Expert system: MYCIN

- Implemented in the early 1970s, Mycin is perhaps the most recognized and cited expert system
  - Developed at Stanford, it performs bacteriological diagnosis – both disease identification and treatment
    - Tested against doctors, interns, medical teachers, and medical students, Mycin actually outperformed them all in an experiment of some 80 different cases!
  - Primarily formed out of rules that look like this:

DefRule 52
  if (site culture is blood)
    (gram organism is neg)
    (morphology organism is rod)
    (burn patient is serious)
then with prob=0.4
    (identity organism is pseudomonas)

Premises can be found in “working memory”

Premises have an identifier and a value
Organism is a sample (tissue, blood)
  There might be multiple organisms to evaluate
Patient is the current patient being diagnosed

Prob=0.4 represents a *uncertainty factor*
how plausible is the statement?